

Region-Routed Metacognition RL (DCPO v2→v4): Hypothesis- Verification Study

Author: autoresearch loop (Claude) **Date:** 2026-06-11 **Project:** metacognition-math, branch ctsd-phase-c **Reviewed sources:** wandb group triobj_dcpo_v3* (runs adf1a0fc v3l, daa6d35a v3m), /tmp/probe_pmi_report.json, /tmp/probe_pmi_cross_report.json, docs/dcpo_v3_design_and_status.md, spec docs/superpowers/specs/2026-06-11-dcpo-v4-likelihood-rmeta-design.md

Executive Summary

Question	Experiment	Result	Verdict
Does a causal counterfactual reward teach the value of meta?	v3 (CF 2nd generation)	acc(with-meta) – acc(without-meta) ≈ 0 on-policy → no learnable gap	signal too sparse
Does the format penalty cause meta mode collapse?	v3l	meta emission 0.5 → 0.0 by step ~60	confirmed (collapse engine)
Does an emission floor prevent the collapse?	v3m (floor 0.1)	collapse delayed 8-12 steps , then total (0 by step 66)	floor = delay, not arrest
Does a dense likelihood reward (PMI) carry signal?	v4 offline probe, n=7,877	real beats placebo t=17.9, $p \approx 10^{-72}$; AUC 0.78	yes, but ...
...is the raw signal content-specific?	cross-shuffle probe	placebo retains 86% , cross-problem meta retains 52%	no — generic-text dominated
Is there a real content signal under the generic component?	placebo-corrected aggregate	right content +0.019 , wrong content –0.046 (vs placebo)	yes, small, directional

Conclusion in one sentence: sparse correctness-difference rewards cannot teach metacognition (no gap to learn from), raw dense likelihood rewards mostly pay for emitting any text, and the trainable signal is the **placebo-corrected content increment** — small, highly significant, and direction-correct — which v4 stage-2 now adopts as its reward.

1. Background and Intent

1.1 What we are trying to do

Meta-CoT is a math-reasoning format in which the model externalizes metacognition inside `<|meta|> ... </meta|>` blocks (self-checks, strategy switches, confidence statements) during its chain of thought. The north-star (CLAUDE.md, 2026-06-07) is: **metacognitive behavior is a means to accuracy, not the goal** — RL must selectively reinforce useful metacognition, with calibration only a sub-goal.

DCPO (our method line) is region-routed RL: each response is partitioned into disjoint token regions — ANSWER (reasoning + final answer), META_CONTENT (text inside meta tags), CONF (the confidence number) — and each region receives its **own** reward head (R_{corr} / R_{meta} / R_{cal} , plus a format head), centered per head with Dr.GRPO group-mean subtraction. Routing is the part that works and survives every redesign below; what changed version to version is **what R_{meta} measures**.

1.2 The central difficulty

R_{meta} must answer: "did this meta block actually help the solution?" Versions differ in how they estimate that:

- **v2:** transition-table proxy (did the answer flip after the meta?) — gameable, replaced.
- **v3 (v3a-v3m):** causal counterfactual — regenerate the same rollout with meta suppressed, reward the correctness difference $c_{\text{with}} - c_{\text{without}} \in \{-1, 0, +1\}$.
- **v4:** dense likelihood (PMI) — how much does the meta block raise the model's own probability of its own subsequent reasoning, $\Delta_t = \log P_{\text{ref}}(C_t \mid \text{prefix} + \text{meta} + C_{(t)}) - \log P_{\text{ref}}(C_t \mid \text{prefix} + C_{(t)})$, scored by a frozen SFT reference at $T=1$ (design adopted from RLT, arXiv:2506.08388, after user feedback).

2. Hypotheses and Verification

H2 — "A causal counterfactual reward provides a learnable signal" → SPARSE (falsified in practice)

The v3 counterfactual machinery itself was made to work: the CF generation initially produced meta-signature echoes that leaked the suppressed meta back into the counterfactual; after signature suppression the gradable CF fraction (dcpo/cw_graded_rate) rose from ~ 0.11 (v3l, steps 11–19) to **0.488** at v3m step 1 (v3m config dcpo_cf_suppress_signature: true; source: wandb runs adf1a0fc/daa6d35a). But the measured quantity was then ≈ 0 : on-policy, **accuracy with meta minus accuracy without meta showed no exploitable gap**, so R_{meta} was a coin flip on most rows and silent on the rest. A reward that is almost always 0 or noise cannot teach why meta helps. This is the direct motivation for a **dense** reward (v4): instead of one ± 1 bit per rollout requiring a second generation, score every continuation token's likelihood shift in one reference forward pass.

H3 — "The format penalty is the collapse engine" → CONFIRMED (v3l)

v3l added a format-violation penalty alongside the meta heads. Meta emission fell from ~ 0.5 to **0.0 by step ~ 60** (Figure 1, blue; wandb run adf1a0fc, dcpo/meta_emit_rate). The mechanism: when emitting meta risks a format penalty but emitting no meta risks nothing, the cheapest policy is silence. The policy found it reliably.

H4 — "An un-centered emission floor prevents collapse" → DELAY, NOT ARREST (v3m)

v3m added dcpo_meta_floor: 0.1: a small positive reward for emitting trusted-class meta, added **after** Dr.GRPO centering (a pre-centering constant cancels in the group mean), per-row normalized (length-neutral), restricted to TRUSTED meta classes.

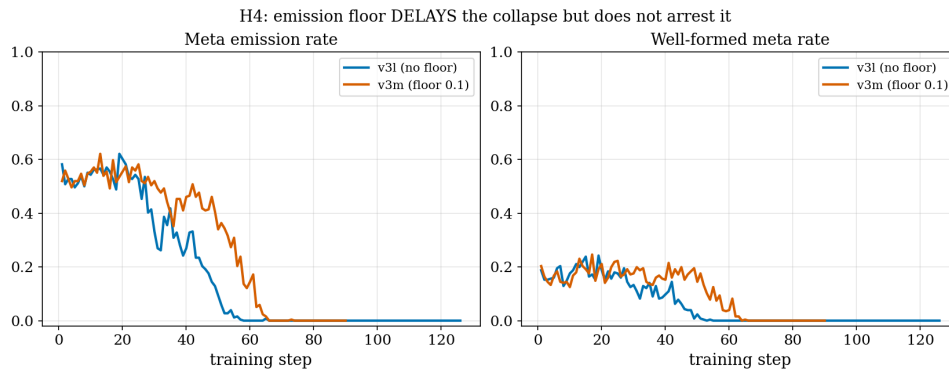


Figure 1 interpretation: both runs start at ~0.5–0.6 emission. v3l (blue) collapses through steps 30–55, falls below 0.05 at step 52 and reaches 0 at step 58. v3m (orange) holds a plateau near 0.4–0.5 longer, falls below 0.05 at step 64 and reaches 0 at step 66 — a delay of **8 steps by first-zero, 12 steps by the 0.05 threshold** (source: wandb `dcpo/meta_emit_rate`, runs `adf1a0fc/daa6d35a`). The well-formed rate (right) tells the same story at lower amplitude. **A floor changes the collapse schedule, not the equilibrium:** as long as the content reward on meta is noise (H2), any competing pressure (format risk, length cost) eventually wins. The fix must make meta content worth something, which is exactly H5–H7.

H5 — "Dense likelihood delta carries decision-relevant signal" → YES (probe, with caveats below)

The v4 kill-or-go offline probe scored 7,877 guard-filtered rollouts (8,024 parsed, from `e8_goldfree_1030_16k_k8`; v8_strict SFT as frozen scorer) in three arms: **real** (the model's own meta at its native position), **placebo** (the meta replaced by a contentless "Let me continue." wrapped in tags), **shuffle** (the meta replaced by another rollout's meta). Three pre-registered kill criteria:

1. **Placebo:** real must beat placebo, paired one-sided t. Result: $t=17.9$, $p \approx 9.6 \times 10^{-72}$ for the mean aggregation (source: `/tmp/probe_pmi_cross_report.json placebo.mean`). **PASS** — decisive across all four aggregations (t 10–52).
2. **AUC:** per-row Δ -aggregate must separate correct from wrong continuations, $AUC > 0.6$ on the entangled split. Result: mean AUC **0.779 overall / 0.696 entangled (n=160) / 0.763 clean** (source: same file, `auc.mean`). **PASS** (Figure 3).
3. **Shuffle:** replacing the meta with someone else's must collapse Δ toward 0 (ratio < 0.25). Result: **FAIL twice** —

and the way it failed is the most informative finding of the study (H6).

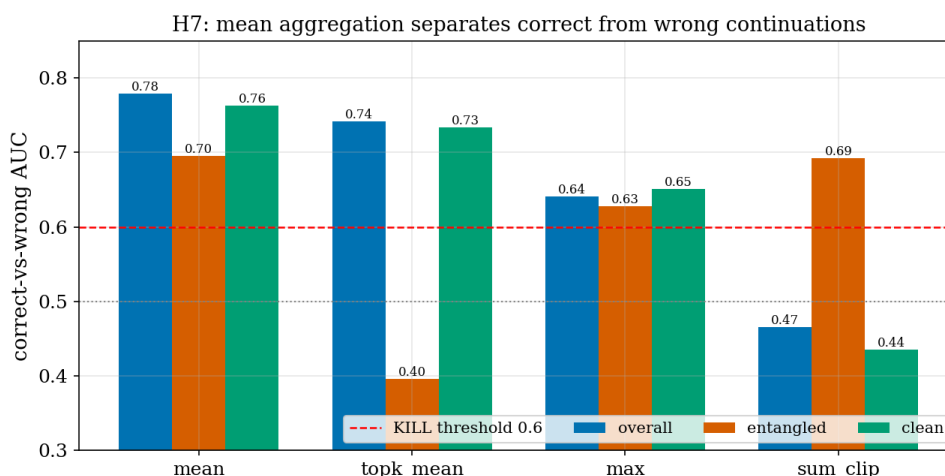


Figure 3 interpretation: mean (per-token average delta) dominates on every split and is the only method above the 0.6 kill line on all three; `sum_clip` is below 0.5 overall (worse than chance — long metas accumulate negative tails). The probe therefore froze `dcpo_pmi_agg: mean, clip_c = 0.342` (p95 of $|\Delta|$) into the stage-2 config (commit 4b169dd).

H6 — "The shuffle failure is a sibling-duplication confound" → HALF-TRUE, and the half that failed is a discovery

First probe shuffled **within problem** (sibling rollouts of the same question): collapse ratio **0.758** — barely collapsed (source: `/tmp/probe_pmi_report.json shuffle.mean.collapse_ratio`). Diagnosis: siblings share most of their solution (measured continuation similarity ~ 0.49 , near-verbatim pairs common), so swapping their metas barely changes content — the criterion, not the signal, looked broken. The disambiguation re-test shuffled **across problems**: collapse ratio fell to **0.516** (source: `/tmp/probe_pmi_cross_report.json shuffle.mean.collapse_ratio`). So the confound is real ($0.76 \rightarrow 0.52$) — but half the signal survives a meta from a different problem, which no content-specific signal should.

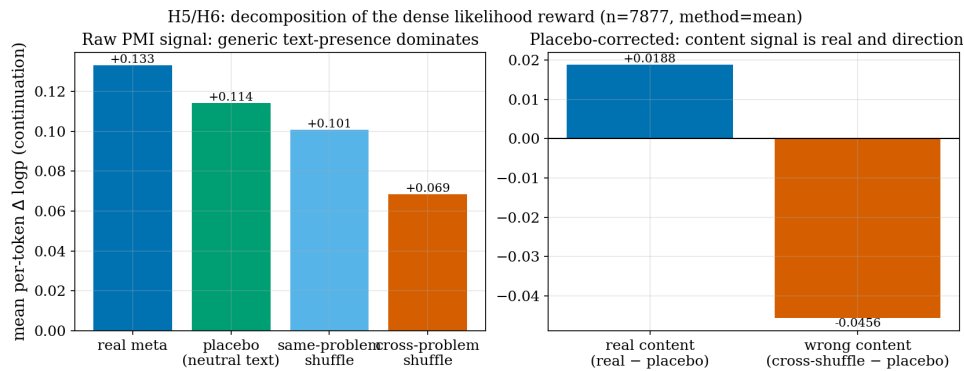


Figure 2 interpretation (left): mean per-token Δ is **+0.133** with the real meta, **+0.114** with a contentless placebo, **+0.101** with a same-problem sibling's meta, **+0.069** with a different problem's meta (sources: `cross report delta_stats.mean.mean`, `placebo.mean.mean_placebo`, `first report shuffle.mean.mean`, `cross report shuffle.mean.mean`). The placebo bar is the discovery: **86% of the raw reward is "some text is present here"** — a generic component that any meta-shaped interruption earns. An RL policy paid with raw Δ would learn to emit meta-shaped anything: exactly the mashing failure v3 fought.

Figure 2 interpretation (right): subtracting the placebo per row isolates the content increment: the right content adds **+0.0188** ($t=17.9$), the wrong content **-0.0456** — worse than saying nothing meaningful. The content signal is small but real and **directional**: it rewards correct-content meta and punishes wrong-content meta, which is precisely the gradient a metacognition reward should carry.

H7 — "The placebo-corrected delta $\Delta' = \Delta - \Delta_{\text{placebo}}$ passes the full battery" → aggregate-level YES; per-row verdict PENDING

At the aggregate level the corrected metric collapses under cross-shuffle by construction of the numbers above ($+0.0188 \rightarrow -0.0456$ is a sign flip, not a mere shrink). The probe was extended (commit 4ad8a09) with a dedicated `verdict_corrected` battery — corrected mean > 0 paired-t, corrected AUC, corrected shuffle where the **shuffle arm is also placebo-subtracted** — plus a per-row dump so any future metric variant can be graded without the ~95-minute GPU re-scoring pass. The per-row corrected probe is running; the open risk is the corrected **AUC**: subtracting placebo removes a component that may itself correlate with correctness, so AUC could drop below 0.6. Decision rule pre-registered: PASS → stage-2 ships

Δ' ; corrected-AUC FAIL $\rightarrow$ stop and re-decide (raw Δ + group-centering argument vs. stage-1-only).

3. Implementation Structure (what stage-1/2 actually run)

```
response tokens: <think> ... <|meta|> META_CONTENT (CONF) <|/
meta|> ... ANSWER
mask partition:  ANSWER_REGION | META_CONTENT\CONF | CONF |
FORMAT_* (disjoint, asserted)
reward heads:    R_corr(w=1.0) $\rightarrow$ ANSWER    R_meta(w=0.5, stage-2
only) $\rightarrow$ META\CONF
                R_cal Brier(w=0.3, stage-2) $\rightarrow$ CONF
R_format(w=0.1) $\rightarrow$ FORMAT
centering:       Dr.GRPO per-head group-mean subtract over
member rows
floor (v3m, kept): +0.1 un-centered AFTER centering, row-
normalized, trusted classes
```

Stage-1 (triobj_dcpo_v4_stage1_h100_4x4k, **launched 2026-06-11, running**): R_corr + floor + R_format only —
dcpo_rmeta_source: none. 50 steps, ckpt every 10. Exit gate:
wellformed $\rightarrow$ 70-80%, emission healthy (no Figure-1 collapse),
 Δ -distribution non-degenerate on the gs50 checkpoint.

Stage-2 (config frozen, **gated** on H7): dcpo_rmeta_source: pmi,
dcpo_pmi_agg: mean, dcpo_pmi_clip_gate: 0.342, sign gate R_meta,row
= (± 1 by correctness) $\times$ clip(mean Δ' , 0, c), CONF carved out of
the dense reward (Brier only), w_meta warmup 50 steps.

```
# v4 dense reward core (src/training/dcpo_pmi.py) – both arms
tokenized
# independently, continuation re-matched by byte identity,
then:
delta_t = logp_with[t] - logp_without[t]      # per
continuation token
raw      = aggregate(delta_t, method="mean")    # probe-frozen
r_meta   = sign * np.clip(raw, 0.0, clip_c)    # sign = +1 if
correct else -1
# H7 amendment (pending corrected-probe PASS): raw  $\rightarrow$  raw -
raw_placebo
```

4. Proposed Improvements (report → plan/ code deltas)

4.1 Ship Δ' (placebo-corrected) as the stage-2 reward — in progress

Problem: 86% of raw Δ is generic text-presence (Figure 2 left); training on it reinforces mashing (the Figure-1 failure mode with the sign flipped). **Solution:** third reference-scoring arm (prefix + placebo + continuation) per meta row; $\Delta' = \Delta - \Delta_{\text{placebo}}$; knob `dcpo_pmi_placebo_correct: true`; clip from `recommendation_corrected`. **Cost:** ref scoring 2→3 forwards per row ($\times 1.5$). **Expected:** reward now pays the content increment (+0.019 right / -0.046 wrong), making "emit empty meta" strictly unprofitable vs. silence — addressing both the H3 collapse pressure and the mashing risk at once.

4.2 Stage-1 exit gate must inspect the corrected Δ distribution

Problem: the pre-registered gate (" Δ -distribution non-degenerate at gs50") was written before H6; a healthy-looking raw- Δ histogram can be pure generic component. **Solution:** run the (already extended) probe on the gs50 checkpoint and gate on `verdict_corrected + corrected  $\Delta'$  stats`; re-freeze `clip_c` from that run (scorer-lineage note in the config).

4.3 Entangled-split coverage is dangerously thin

Problem: the load-bearing AUC split has $n=160/7,877$ (2%) — the probe's discriminative claim rests on few rows (Figure 3 caption n). **Solution:** log the entangled fraction during stage-2 training (`dcpo/entangled_frac`) and re-run the probe battery on stage-2's own rollouts at first checkpoint; if entangled coverage stays $\sim 2\%$, the per-problem AUC claim needs widening (e.g., relax the echo-detection n-gram from 8).

4.4 Keep the floor, stop trusting it

Problem: H4 showed the floor only buys schedule. **Solution** (already in configs): keep floor 0.05–0.1 in stage-2 as a transition aid while Δ' becomes the load-bearing incentive; watch for the Figure-1 signature in the first 60 steps of stage-2 specifically — if emission decays while Δ' rewards flow, the

content signal is too small at $w_{\text{meta}}=0.5$ and the warmup schedule (M4) should be extended rather than the floor raised.

5. Limitations

- **Frozen-scorer lineage:** the probe scored with $v8_{\text{strict}}$ SFT; stage-2 entry policy is stage-1's $gs50$. Numbers ($clip_c$, AUC) must be re-frozen on the $gs50$ probe (4.2) — current values are a baseline, not a guarantee.
- **Aggregate vs per-row:** H7's sign flip is aggregate-level; the per-row corrected AUC (the actual kill criterion) is pending. A placebo-correlated correctness component could still sink it.
- **Cross-problem shuffle is off-distribution:** a different problem's meta is maximally wrong content; the $0.52 \rightarrow \text{collapse}$ claim may overstate how much a near-miss wrong meta is punished.
- **Entangled split thinness** (4.3): $n=160$ — the population the method most needs to win on is the one least measured.
- **One-seed trajectories:** Figure 1 compares single runs per condition ($v3l/v3m$); the delay magnitude (8–12 steps depending on threshold) is not error-barred, though the 0-equilibrium endpoint is unambiguous in both.

6. Next Experiments

E-s1: v4 stage-1 (RUNNING, triobj-dcpo-v4-s1)

- **Tests:** format/emission stabilization without content reward (pre-condition for H7's stage-2).
- **Exit gate:** wellformed 70–80%, no Figure-1 collapse signature, corrected- Δ probe on $gs50$ (4.2).

E-corr: per-row corrected probe (RUNNING, local A100, verdict $\approx$ 00:50)

- **Tests:** H7 per-row. **Expected if H7 true:** corrected shuffle ratio $\ll 0.25$ (sign flip), corrected AUC $\geq \text{raw}-0.05$. **Falsified if:** corrected entangled AUC $< 0.6 \rightarrow$ pre-registered stop + re-decision.

E-s2: v4 stage-2 with Δ' (GATED on E-s1 $\wedge$ E-corr)

- **Config changes** (vs frozen stage-2 yaml):
yaml
dcpo_pmi_placebo_correct: true # new knob (4.1)
dcpo_pmi_clip_gate: <from recommendation_corrected @ gs50
probe> # was 0.342 (raw, v8_strict)
- **Expected:** meta emission survives (unlike Figure 1)
because content-bearing meta now out-earns silence;
calibration (R_cal Brier) unaffected by the carve-out; final
metric = accuracy delta vs stage-1-only on the 1,030-
problem eval.

7. Conclusion

Across v2→v4 the constant is the region-routing architecture; the variable is the content reward, and each failure sharpened its definition. v3 proved a causal-but-sparse reward starves (H2) and that emission incentives alone only reschedule collapse (H3-H4, Figure 1). The v4 probe proved a dense likelihood reward is statistically loaded (H5) but decomposes into a dominant generic-text component and a small, significant, directional content component (H6, Figure 2) — and that only the latter is worth training on (H7). The study's practical output is already shipped: stage-1 is training, the corrected battery is measuring, and stage-2's reward is re-specified as the placebo-corrected content increment.

References

- RLT: "Reinforcement Learning Teachers of Test Time Scaling" (arXiv:2506.08388) — dense student-likelihood teacher rewards; basis of the v4 two-round + PMI design.
- RLSD/SDPO line (memory `rlsd-vs-sdpo-reference`) — warning that gold-conditioned teachers suppress epistemic content; motivated gold-free v4 scoring.
- Spec: `docs/superpowers/specs/2026-06-11-dcpo-v4-likelihood-rmeta-design.md` (review findings C1-C3/I1-I5/M1-M4).